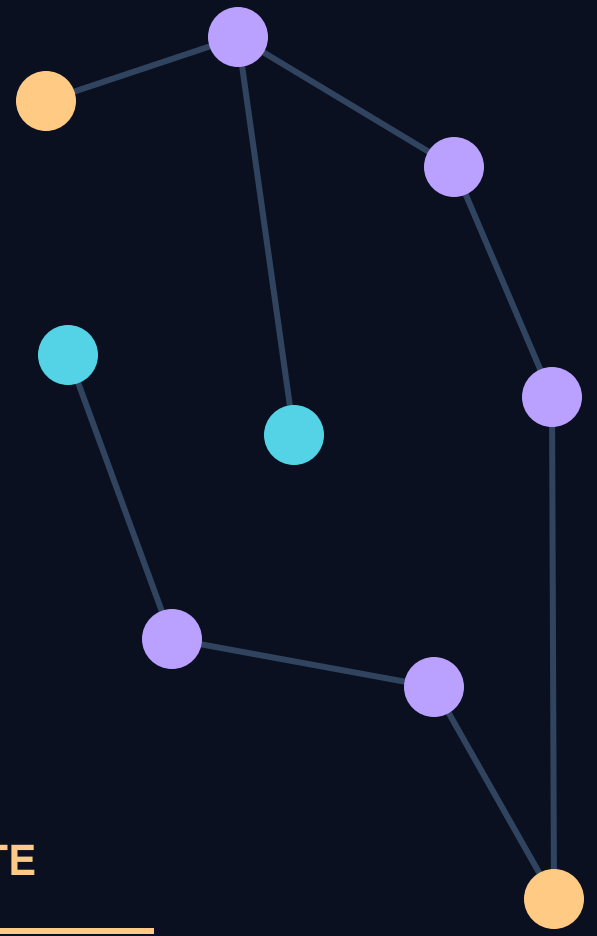
QCCR • RESEARCH PRESENTATION

GEOMETRY-AWARE QUANTUM CLOUD



Getting a better Bell pair

Routing, noise, and the cost of throwing away attempts

01 MAKE

02 BREAK

03 PROTECT

04 INNOVATE

We measure the pair, the work, and the retries

We test a shared quantum state with three measurement settings: XX, YY, and ZZ.

What the measurements should show

Basis	Ideal relationship
ZZ	Equal bits: 00 or 11
XX	Equal bits: 00 or 11
YY	Opposite bits: 01 or 10

Target: $(|00\rangle + |11\rangle) / \sqrt{2}$

$F = (1 + \langle XX\rangle - \langle YY\rangle + \langle ZZ\rangle) / 4$

Fidelity F: similarity to the target pair

1 is ideal. A higher value means the measured correlations are closer to the target.

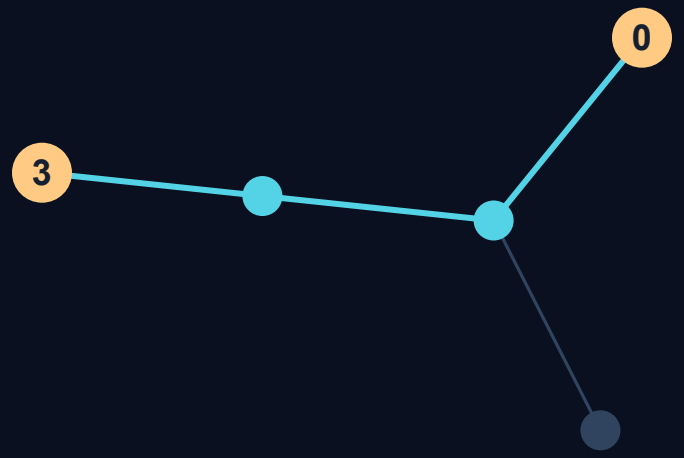
Yield: the fraction of attempts we keep

If 1,000 attempts produce 520 passes, yield is 52%. The other 480 still cost gates.

The fixed endpoints determine the route

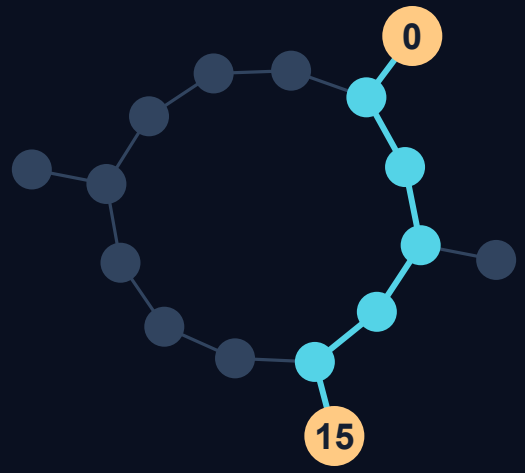
A dot is a physical qubit. A line is an allowed neighbor link. Orange dots are the required output ports.

2016-Inspired (reference)



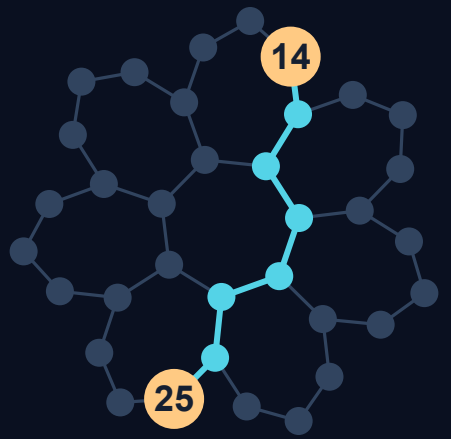
Ports 0 and 3
3 links apart

IBM snapshot



Ports 0 and 15
6 links apart

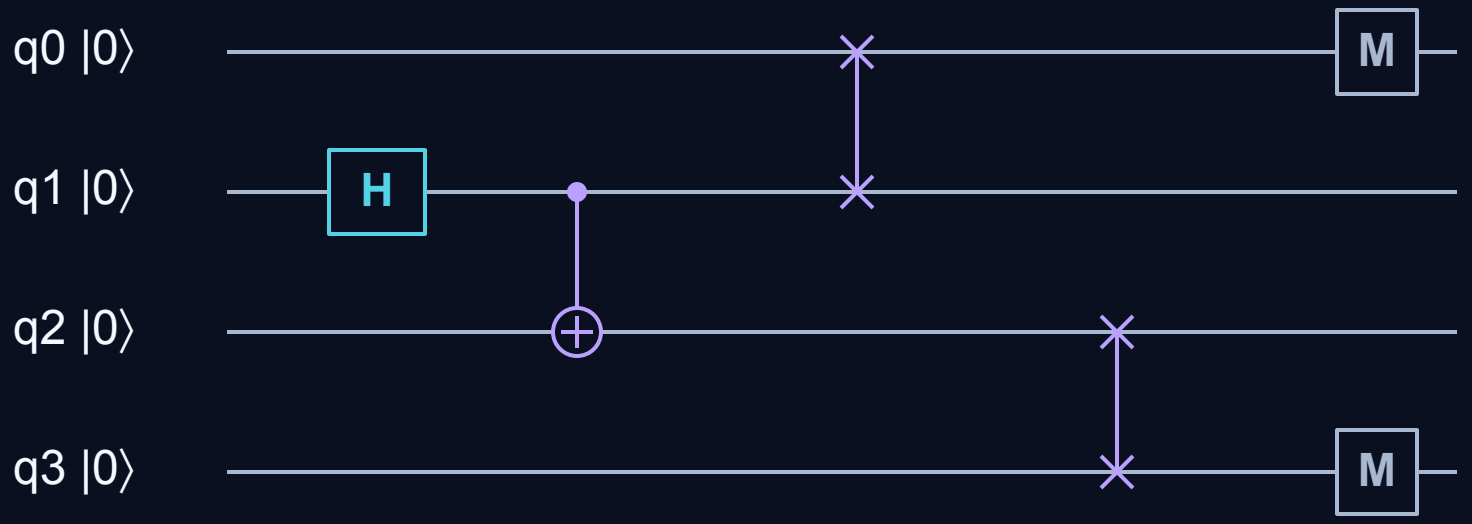
Hyperbolic patch



Ports 14 and 25
7 links apart

Two SWAPs turn a short route into seven CNOTs

2016 example: first make the pair on the middle link, then move each half to an endpoint.



(|00⟩ + |11⟩) / √2

Prepare the Bell pair:

- **H creates superposition**
- **CNOT creates entanglement**

1 + 2 × 3 = 7

CNOT gates after compilation

●—⊕ CNOT
×—× SWAP
M measurement
H creates superposition

Same circuit: F = 1.000 without noise → 0.862 with the official noise

Checking raises the score of the kept sample

MAKE: ideal F = 1.000 → BREAK: add official noise → PROTECT: check parity and keep reported passes.



Kept attempts in Level 3

Model	Kept
Reference	85.7%
IBM snapshot	82.2%
Hyperbolic	72.3%

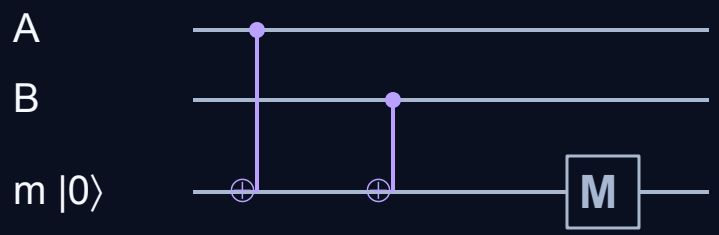
The check selects a cleaner sample while spending more gates and attempts.

The check adds noise; selection removes many bad attempts

Reference circuit: compare the original run, the checked run before filtering, and only its reported passes.



Two CNOTs record the parity



Logical check; hardware routing adds SWAPs.

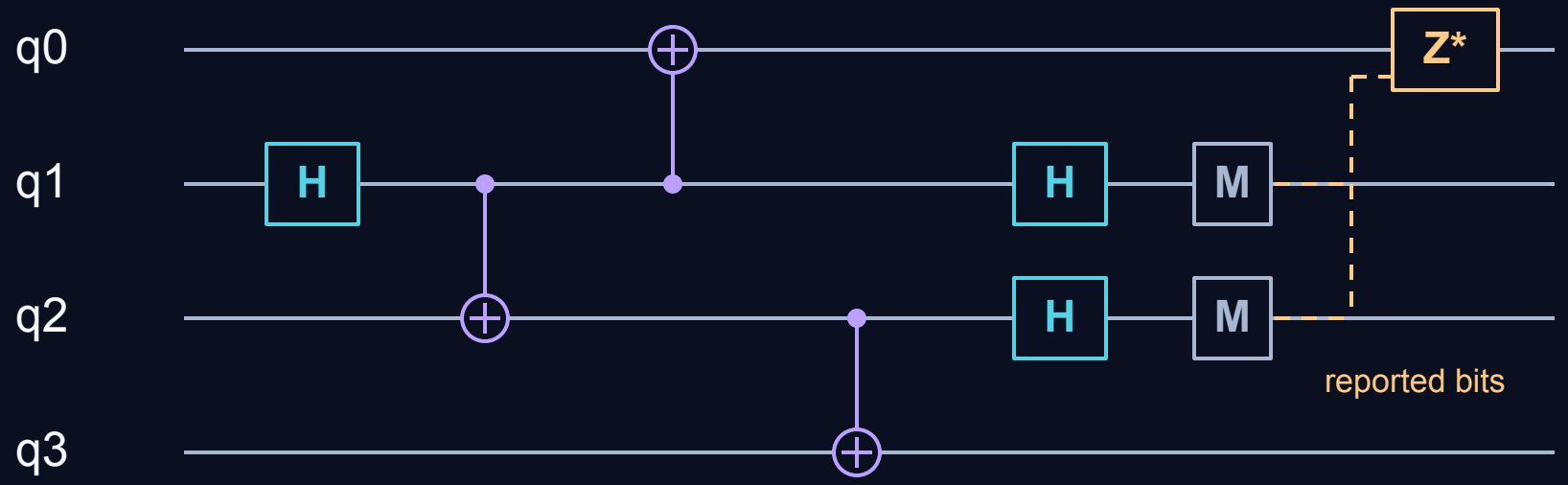
85.7% of attempts accepted

7 → 17.5 CNOTs per accepted result

A phase error can still pass this check. Equal Z bits are only part of the Bell-state test.

Level 4 connects the endpoints by measuring the middle

Build a superposition of 0000 and 1111. Measure the middle, then correct the endpoint sign.



3 CNOTs

All attempts kept

H then M measures X.
*Apply Z if the two reported bits differ.

Shared four-qubit state → two-qubit Bell pair

00 or 11 → no phase correction **01 or 10 → apply Z**

Z fixes a possible minus sign between the Bell-state terms, 00 and 11.

Fewer gates help on both synthetic routes

Corrected GHZ keeps every attempt. The IBM result stays about the same despite the large gate saving.



CNOTs / attempt	Reference	IBM snapshot	Hyperbolic
SWAP → GHZ	7 → 3	16 → 6	19 → 7

IBM readout errors offset the saving in gates

We reran the same circuits while turning noise components on separately.



GHZ saves noisy gates, but it relies on correctly reading the middle qubits.

Level 5 keeps attempts where neighboring data qubits agree

Spread one Bell half across neighboring data qubits. Check agreement, then decode it back to one output.

Encode
one data qubit → three

Check neighbors
same / different

Decode
three data qubits → one

Expected out of 1,000	Data pattern	Check result
512: no bit flips	000 and 111 remain linked	Keep: desired Bell pair after decoding
480: some bits flip	Neighbor disagreement	Reject
8: all three flip	All still agree	Keep: bit-flipped Bell pair after decoding

Each of the three data qubits has an independent 20% chance of a storage bit flip.

520 kept attempts

512 / 520 = 0.985 fidelity

A group that agrees is much more likely to have no flips than to have every bit flipped.

Three data qubits reach the example 0.95 target at lower cost

More data qubits suppress all-flip passes, but add noisy gates and reduce the fraction kept.

Data qubits	Fidelity [95% interval]	Kept	CNOTs / kept shot
1	0.794 [0.788, 0.800]	100.0%	1.00
2	0.929 [0.924, 0.933]	67.7%	7.38
3	0.972 [0.968, 0.975]	51.7%	17.42
7 (path)	0.978 [0.973, 0.983]	20.5%	122.00
7 (closed cycle)	0.975 [0.970, 0.980]	20.8%	129.78

Independent confirmation

F: 0.795 → 0.972 | 51.8% of attempts kept

Meet the fidelity target with the least gate work

Have the cloud compiler choose the protocol using fidelity and the cost of retries.

Delivery

Try corrected GHZ on the synthetic routes.

It improves fidelity with fewer CNOTs and keeps every attempt.

IBM readout

Improve intermediate measurements before relying on GHZ's gate savings.

A wrong reported bit can select the wrong correction.

Bit-flip storage

For the tested 0.95 target, use three data qubits.

F ≈ 0.972; keep ≈ 52%; about 17.4 CNOTs per kept shot.